AI Ethics

Course Num: SEIS 651

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Week 9

Agenda

Focus Area

AI + Ethics + Individuals: Fairness

This is my favorite week because…

This week is about practically achieving equitable human impact!

Week 9

Outline – Key Points

- Measuring fairness
- Balance with other ethical and practical considerations
- Accuracy vs. fairness
- Fairness in practice
- Responsibility vs. innovation

Lecture

Recap: Week 8

- AI in Context – how is it being used?
- Determine appropriateness of reflecting bias in your model's output (what's the **goal**?)
- Consider bias in **input**, **output**, and **impact**
- **Collaborate** with people who can see bias differently than you!
  - (Remember: it's unlikely that any one person is completely without bias, or can recognize all potential sources and types of bias!)
- *Continually assess your own biases, standards, assumptions, and perspectives, and critically evaluate your work!*

Defining Fairness

There are several methods for measuring and assessing fairness. Examples include:

- "Statistical Parity":
  - No significant difference in predictions by class/group
    - *E.g., same percentage of men and women receive job offers*
    - *E.g., if Statistical Parity Difference = 0*

- "Equal Odds":
  - No significant difference in prediction by class/group *given an outcome*
    - *E.g.,True Positive and True Negative rates are the same*
  - Requires "equal errors": even distribution of errors by class/group
    - *E.g., False Positive and False Negative rates are the same*

Can we address all of this with the same model?

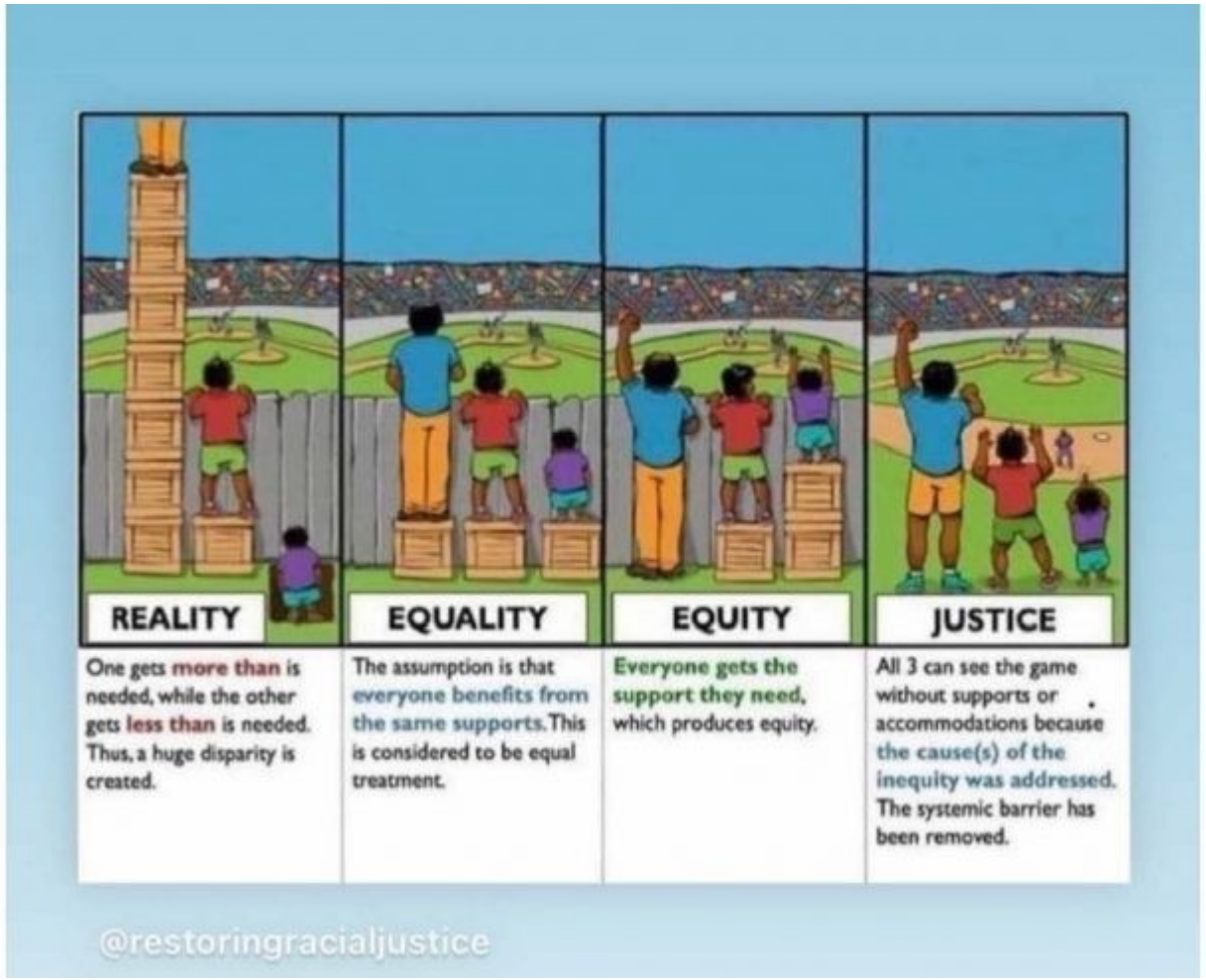
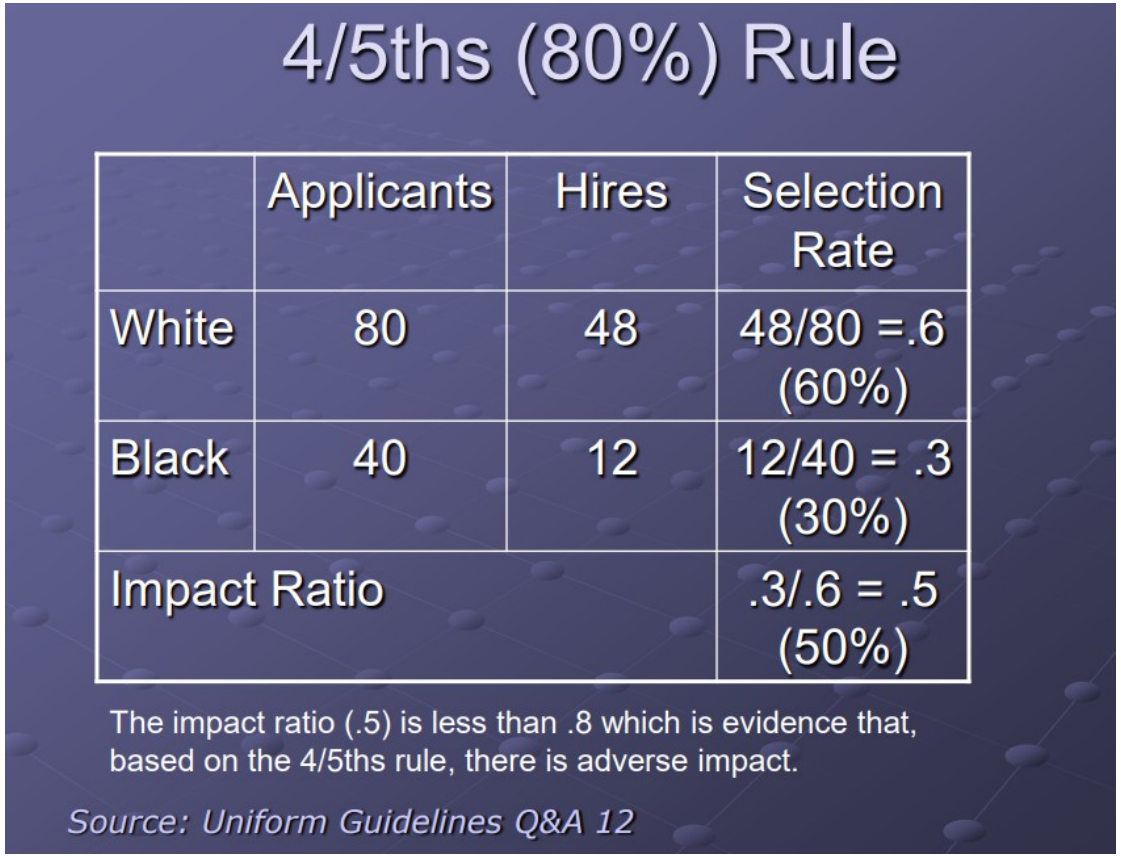


Image credit: https://www.bu.edu/diversity/resource-toolkit/inequity-equality-equity-and-justice/ from cited Instagram creator
**There are MANY variations of this image

Measuring Fairness - Examples

- **Assess for Adverse Impact**
  - Example: 80% Rule for Impact Ratio
  - http://annex.ipacweb.org/library/conf/08/brink.pdf *(note, in this document, "AI" stands for "Adverse Impact")*
- **Statistical Significance testing**
  - Example: compare control and treatment groups, or null and alternative hypotheses
- **Evaluate accuracy by class/group**
- **Evaluate unique aspects of each class/group**
  - Example: unique words in text

- **Additional Resource:** AIF360 documentation *https://aif360.readthedocs.io/en/latest/modules/sklearn.html (note, some of the AIF360 sites have outdated certificates…)*



Balancing Fairness With…

Privacy

- Simply excluding sensitive data, demographics, or other characteristics does not automatically ensure fairness
  - Why not?
  - What about privacy?

Reality

- What happens when reality is biased?
  - Complexity in balancing “neutrality” in our data with the risk of cultural erasure? (e.g., overcorrecting in LLMs or image/audio/video generation)
- Does equal opportunity satisfy a requirement for fairness?

Balancing Fairness With…

Fairness?!

- Some measures of fairness are incompatible with other measures of fairness (statistical parity vs. distribution of errors)

- What should you use?!

- Consider disparate **treatment** as well as disparate **impact**

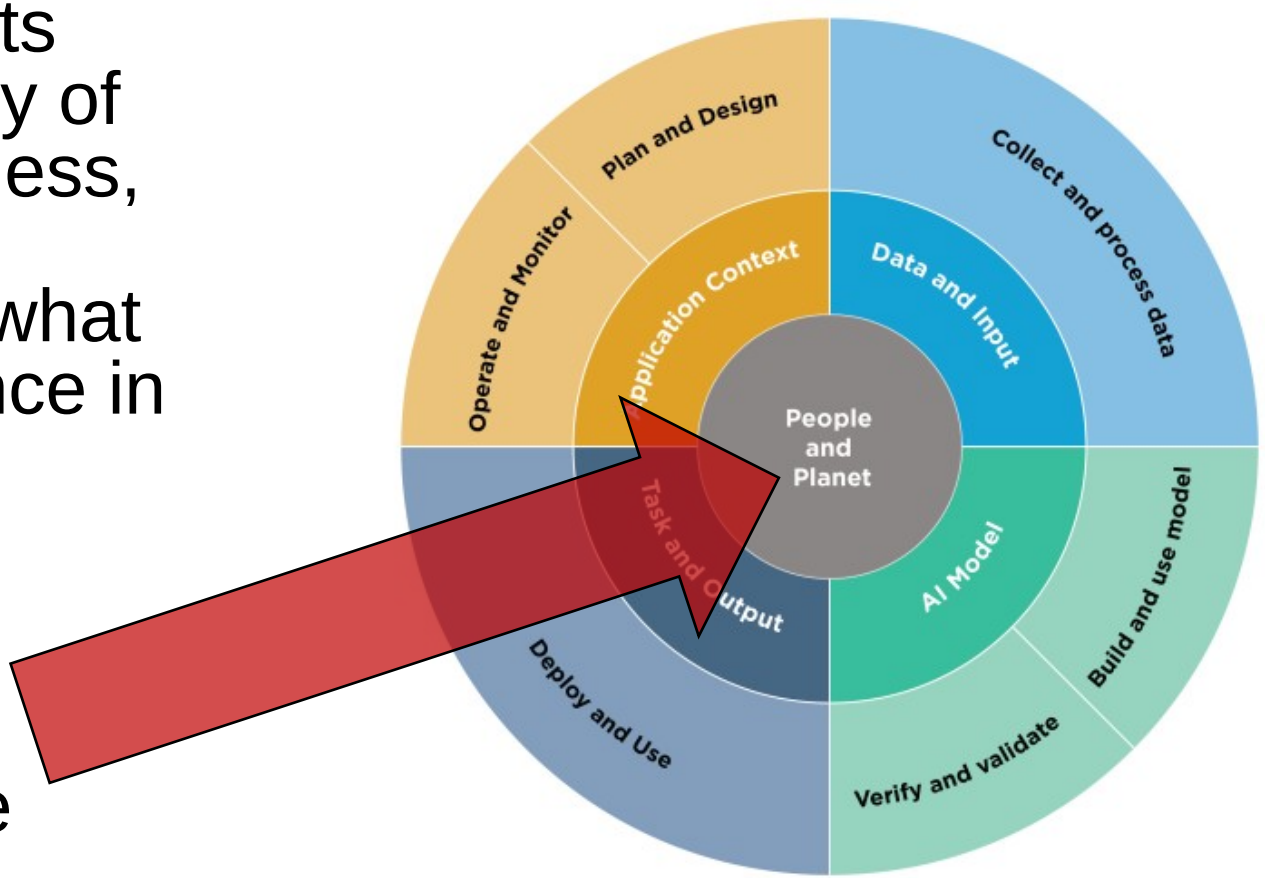
Fairness vs. Accuracy

- Does optimizing for fairness by definition reduce accuracy? (“accuracy/fairness tradeoff”)
  - Sometimes!
- Is there still a tradeoff after you balance or debias your training data?
  - Sometimes!
- So what happens if you really can’t have both?

Challenges In Practice

- “Practically, data scientists have the ethical necessity of considering societal fairness, despite there not being universal consensus on what that means.” (Data Science in Context, p.195)

- Fairness in practice is a concern beyond AI. Remember what is at the center of our AI RMF…



Case Study

- Turnitin's AI detector AI Writing Detection (turnitin.com)
  - A recent iteration **claimed** to have less than a 1% false positive rate.
  - But is incorrectly flagging 1 out of every 100 or even 1,000 human-written documents acceptable when someone's grade and reputation are at stake?
  - Beyond the fairness and accuracy of the model itself, is the way in which we use this model fair to those it will impact?
    - How can we ensure fairness in practice beyond fairness in the model?

Responsibility vs. Innovation

- Consider advertisements vs. the real thing:
  - Just because a company *says* their model can do a thing doesn't mean their model can actually do the thing.

- Which burger do you want to serve a customer? Which one do you want to eat?
  - **Is there a tradeoff here?**



Image: https://www.businessinsider.com/fast-food-ads-vs-the-real-thing-2012-1#he-even-asked-for-these-whoppers-to-be-made-specifically-like-the-ones-in-the-ads-they-didnt-turn-out-much-better-3

Responsibility vs. Innovation

Consider…

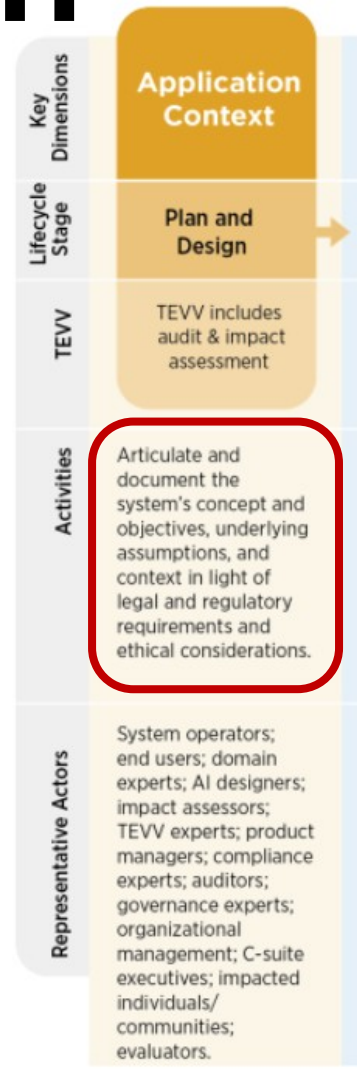
- Why is it so easy to see biased results in generative AI platforms?
- Why is it so easy to find examples of predictive models that demonstrate disparate impact?
- Why are models released with clear and present issues?

Responsibility vs. Innovation

What can we do?

- Define your requirements for model release
  - Threshold for fairness?

NIST AI RMF



- Evaluate: Who does the model work for? Who benefits? Who does it fail for? What type of errors are you seeing?

- *"Data scientists: we should not be the arbiters of truth. We should be translators of ethical discussions that happen in larger society." -Cathy O'Neil*

Recap: Week 9

- Fairness is not universally defined. And some measures of fairness may conflict with other measures!
- Assess model fairness in the context of other ethical and practical considerations (e.g., privacy, history, etc.)
- There may be a tradeoff between fairness and accuracy, but not always! Consider optimizing for both where possible.
- Consider disparate treatment vs disparate impact
  - Disparate impact is still possible even when there is no overt disparate treatment
- Fairness is an important concern in and beyond the model. (AI in context!)
- Responsibility vs. innovation (i.e., does your model just *look good* or is it actually good)

References

- The Ethical Algorithm (Book)
- Data Science in Context (Book)
- Fairness Criteria through the Lens of Directed Acyclic Graphs: A Statistical Modeling Perspective
- Fair prediction with disparate impact: A study of bias in recidivism prediction instruments
- Equality of Opportunity in Supervised Learning
- Is There a Tradeoff Between Fairness and Accuracy?
- Evaluation and improvement of algorithmic fairness in COVID-19 severity classification using Explainable Artificial Intelligence-based bias mitigation (Nejadshamsi et al)